

Exercise A

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Let's consider the generic Markov Decision Process (MPD) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a generic policy π .

- Provide the pseudo-code for the Iterative Policy Evaluation algorithm to estimate V_π in the generic MDP; in the algorithm consider
 - having reached K iterations as unique exit condition;
 - don't report the 'in-place' version of the algorithm.
- Consider the deterministic small gridworld reported in the Figure above: the action space is $\mathcal{A} = \{\text{up, down, left, right}\}$ and if an action takes the agent off the grid, the state remain unchanged. The state space is $\mathcal{S} = \{1, \dots, 9\}$ and the terminal states are reported in grey. All actions are associated, no matter the state they are taken from, with a reward equals to -1. The state transitions are the obvious ones. Consider a deterministic policy π , ie:

$$\pi(\text{up}, s_j) = 1 \quad \forall s_j \in \mathcal{S}$$

$$\pi(\text{left}, s_j) = \pi(\text{right}, s_j) = \pi(\text{down}, s_j) = 0 \quad \forall s_j \in \mathcal{S}$$

Estimate V_π by using iterative policy evaluation with:

- $\gamma = 1$;
- initial estimation $V_\pi = (0, 0, 0, 0)$;
- don't consider the 'in-place' version of iterative policy evaluation;
- stop the algorithm after reaching $K = 3$ as exit condition.

Report the procedure with calculations.

- Suppose to apply Policy Improvement after k steps of the policy evaluation procedure described above. Report the obtained policy after $k = 1, \dots, 3$ steps.

Question 1

In the context of the $TD(\lambda)$ algorithm for model free prediction:

- define the n -step return $G_{t:t+n}$ and the corresponding n -step temporal difference learning updating step for the value function;
- define the λ -return G_t^λ and the corresponding Forward-view $TD(\lambda)$ updating step for the value function;
- describe the Backward-view $TD(\lambda)$ algorithm;
- is there a case where Forward and Backward views are equivalent?

Question 2

Consider the full Reinforcement Learning scenario.

- Report the pseudo-code for the Q-Learning algorithm;
- Report the pseudo-code for the SARSA algorithm;
- Comment on the differences between Q-learning and SARSA and provide the backup diagrams for the two algorithms;
- Report the pseudo-code for the Expected SARSA algorithm and its backup diagram;
- Comment on the differences (pros and cons) between SARSA and Expected SARSA.

REPORT IN THE FOLLOWING THE TOTAL NUMBER OF PAGES YOU ARE SUBMITTING. IN EACH PAGE YOU MUST INCLUDE YOUR NAME/SURNAME AND YOU MUST ENUMERATE EACH PAGE INCREMENTALLY.